Md Ashaduzzaman

US Permanent Resident | Greater Boston, MA | Contact number: (531)-239-7157 | Email: asadayon@gmail.com
LinkedIn: in/md-ashaduzzaman-ma | GitHub: asadayon | Scholar: Md Ashaduzzaman

Summary

• 3+ years of experience developing **LLM** and **AI-Chatbot** applications, including Q&A tools with embeddings and **RAG**, and published work on recommendation-specific answer tailoring.

• 4+ years of experience in **computer vision** for medical image analysis, specializing in **self-supervised** learning and **fine-tuning** foundational models for precise **segmentation** of microstructures in Cryo-TEM and SEM images.

• Specializing in the intersection of **Generative AI** and Human-Computer Interaction (**HCI**)—specifically, how we can leverage conversational agents to make recommendations more transparent and comprehensible for users.

Skills

Programming & ML	LLM	Data Engineering	MLOps & Infra
Python (NumPy, Pandas, scikit-learn), Git, SQL, PyTorch, TensorFlow, LangChain, Hugging Face	LLM Fine-Tuning, LoRA/PEFT, RAG, AI Agent, RLHF, Safety & Ethics (Bias Detection, Guardrails)	Preprocessing, Feature Engineering, Synthetic Data, ETL, SQL/NoSQL	AWS, Docker, FastAPI, MLflow, Ollama, Streamlit

Education

- PhD in Computing and IT, University of Nebraska Omaha, Omaha, NE, USA *[Expected Graduation, Jul 2026]*
  Concentration on Artificial Intelligence, CGPA: 3.80/4.00
- Master of Science, University of Nebraska Omaha, Omaha, NE, USA *[Dec 2023]*
  Major in Computer Science (Computer Vision), CGPA: 3.76/4.00

Work Experience

Data Science Summer Intern at **Pfizer**, Groton, CT, USA *[Jun 2023 – Aug 2023]*

- Develop an **AI-based Question & Answering tool** using RAG-based LLM, ensuring the confidentiality of Pfizer's private information, and enhancing information retrieval and decision-making processes.
- Used T5 LLM for question answering, LangChain for AI Agent, Sentence Transformer for embeddings, Python, ipywidgets for visualization, Hugging Face for model downloads, FAISS for vector indexing, AWS S3 for storage, and GitHub for version control and code management.

Graduate Research Assistant at University of Nebraska Omaha, NE, USA. *[June 2024 – Present]*

- Utilized LLMs to enhance explanations for a recommender system in an **NSF** project on Empowering Human-AI Collaboration through Conversational Explanations.

Research Projects

- Developed an **explainable** architecture for **recommender systems** using **LLMs**, prompt engineering, advanced NLP and machine learning techniques to deliver personalized, context-aware natural language explanations. [Github] [UMAP] [Springer]
- Performed **semantic segmentation** of Lipid Nanoparticles from Cryo-TEM images using **self-supervised** learning techniques and automatic LNPs size distribution quantification. Link: [Nature] [Github]
- Improved the **segmentation** IoU of SEM Biofilm images using Deep Learning **Super-Resolution**. Link: [Github] [IEEEXplore]
- Developed an **AI-based** framework for **detecting** cells and byproducts in low-volume SEM images of Biofilms. Link: [Github] [frontiers]
- Analyzed and interpreted particle size distribution and porosity (pore size and density) measurements on microspheres SEM images using python libraries (cv2, skimage, scipy, matplotlib) and MATLAB. Link: [Github] [ELSEVIER]
- Performed data mining techniques [FP-tree, outliers] to **detect fraud** in admission exams. Link: [IEEEXplore]

Accomplishments and Awards

- Awarded **"Honorable Mention"** for poster-BYTES session at the C-BYTES Research Showcase, 2024, UNO.
- Contributed to peer-reviewed 4 journals, 9 international conference papers, and 1 book chapter.
- Awarded 2021-22 Graduate Research and Creative Activity (GRACA) **grant**.

Activities and Leadership

Vice President, Bangladesh Student Association (BSA), UNO. *[Aug 2024 – May 2025]*

- Co-founded and led the establishment of the BSA, managing official documentation and collaborating with team members to successfully organize campus events, including the Freshers' reception.

Graduate Teaching Assistant at University of Nebraska Omaha, NE, USA. *[Jan 2021 – May 2024]*

- Facilitated lab sessions, providing hands-on experience in Python and Java programming, while grading assignments, quizzes, and exams to assess student comprehension, and offering mentorship during office hours, enhancing instructional and mentoring skills.

Student Mentor at UNO *[Summer 2024, 25]*

- Mentored 5 high school interns at UNO, guiding RAG-based chatbot development; one project accepted for an international conference poster. [DG.o]
- Supervised 4 undergraduates under the UR2PhD program, earning CRA certification and applying culturally responsive mentorship practices. [Certificate]

Selected Publications

- **Ashaduzzaman, Md**, and Chun-Hua Tsai. "Leveraging Generative AI to Improve Comprehensibility in Social Recommender Systems." *Adjunct Proceedings of the 33rd ACM Conference on User Modeling, Adaptation and Personalization*. 2025.
- **Ashaduzzaman, Md**, Thi Nguyen, and Chun-Hua Tsai. "Explaining Social Recommendations Using Large Language Models." International Conference on Disruptive Technologies, Tech Ethics and Artificial Intelligence. Cham: Springer Nature Switzerland, 2024.
- **Ashaduzzman, Md**, et al. "Using deep learning super-resolution for improved segmentation of sem biofilm images." *2022 IEEE International Conference on Bioinformatics and Biomedicine (BIBM)*. IEEE, 2022.
- **Ashaduzzaman, M.**, & Subramaniam, M. (2023). Self-Supervised Learning-Based Classification of Scanning Electron Microscope Images of Biofilms. In *Machine Learning in 2D Materials Science* (pp. 109-132). CRC Press.
- Bommanapally, V**., Ashaduzzman, M**., Malshe, M., Chundi, P., & Subramaniam, M. (2021, December). Self-supervised learning approach to detect corrosion products in biofilm images. In *2021 IEEE International Conference on Bioinformatics and Biomedicine (BIBM)* (pp. 3555-3561). IEEE.
- Abeyrathna, D**., Ashaduzzaman, M**., Malshe, M., Kalimuthu, J., Gadhamshetty, V., Chundi, P., & Subramaniam, M. (2022). An ai-based approach for detecting cells and microbial byproducts in low volume scanning electron microscope images of biofilms. *Frontiers in Microbiology*, *13*, 996400.
- Islam, F., Das, S**., Ashaduzzaman, M**., Sillman, B., Yeapuri, P., Nayan, M. U., ... & Kevadiya, B. D. (2024). Development of an extended action fostemsavir lipid nanoparticle. *Communications biology*, *7*(1), 917.

Certification

- DeepLearning.AI: Fine-tuning and Reinforcement Learning for LLMs [Credentials]
- DeepLearning.AI: Deep Learning Specialization Certificate. [Credentials]
- Build a Neo4j-backed Chatbot using Python. [Credentials]